DURGA SANDEEP NAMBALLA

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PROFESSIONAL SUMMARY

M.S. in Computer Science (DigiPen, 2026) specializing in real-time rendering, seeking a PhD in computer graphics. Implemented physically based shading, HDR image-based lighting, deferred shading, moment shadow mapping, and SSAO in C++/OpenGL/GLSL. Brings 3 years of professional game development in Unity/C# and 5+ years as an Assistant Professor teaching game programming and mentoring student teams.

EDUCATION

DigiPen Institute of Technology — Seattle, USA
Master of Science in Computer Science — Sep 2024 – Apr 2026

Relevant coursework: Advanced Computer Graphics, Advanced Real-Time Rendering, Advanced Animation & Modeling I, Curves and Surfaces, Machine Learning, Deep Learning

Jawaharlal Nehru Architecture and Fine Arts University — Hyderabad, India
Bachelor of Science (Computer Science and Game Development) — Aug 2011 – Mar 2015

RESEARCH INTERESTS

Real-time rendering, physically based rendering, global illumination

ACADEMIC PROJECTS

Course Projects – Computer Graphics & Game Engines

• Advanced Computer Graphics (OpenGL, C++) [GitHub]

- Implemented the core 3D rendering pipeline including homogeneous coordinate transformations and perspective projection.
- Developed a physically based GLSL microfacet BRDF lighting model incorporating diffuse, specular, and Fresnel effects for realistic shading.
- Integrated key rendering techniques including texture mapping, sky dome rendering, normal mapping, and dual-paraboloid environment reflection mapping.
- Designed and optimized a two-pass shadow mapping pipeline using FBOs and shader-based techniques and built HDR image-based lighting (IBL) and irradiance mapping workflows to enhance visual realism.

• Advanced Real-Time Rendering (C++, OpenGL, GLSL) [GitHub]

- Built a deferred shading renderer with a multiple-render-target G-buffer, supporting many local lights rendered as additively blended bounding volumes with smooth range-limited attenuation.
- Implemented Moment Shadow Mapping (Peters & Klein, I3D 2015) and Variance Shadow Maps (Donnelly & Lauritzen, I3D 2006) with separable Gaussian blur in compute shaders, and compared light leaking across the methods.
- Replaced the fixed-width blur with summed-area tables (Crow, SIGGRAPH 1984), allowing variable-size shadow filtering in constant time per pixel.
- Implemented HDR image-based lighting with a microfacet BRDF: precomputed diffuse irradiance maps (multithreaded CPU integration) and Monte Carlo importance-sampled specular reflection (Colbert & Křivánek, GPU Gems 3), with tone mapping and linear-to-sRGB output.
- Implemented Alchemy screen-space ambient obscurance (McGuire et al., HPG 2011) with an edge-aware bilateral filter, applied to ambient lighting only.

• Advanced Animation and Modeling I (OpenGL, C++) [GitHub]

- Implemented 3D animation and modeling techniques in C++ using OpenGL, including hierarchical character animation, quaternion-based transformations, interpolation algorithms, and smooth path-following motion.
- Developed physically based simulation and inverse kinematics systems using forward/inverse kinematics, rigid-body dynamics, collision response, and physics-based modeling for real-time applications.

• Curves and Surfaces (JUCE framework, Unity) [GitHub]

- Developed an interactive computational geometry application implementing Bezier Curves, B-Splines, De Casteljau, De Boor, Newton Interpolation, and Cubic Spline algorithms with real-time curve rendering and visualization.
- Built an interactive editor supporting dynamic control-point manipulation, knot vector management, and multiple curve evaluation techniques, enabling real-time visualization of polynomial, spline, and parametric curve behavior.

• 2D Game Engine (C++, OpenGL) [GitHub]

- Collaborated within a 4-member team to build a custom game engine using C++ and OpenGL, designing UI systems, debugging tools, and contributing to the rendering pipeline.
- Developed and evaluated three gameplay prototypes, selecting the most viable concept and delivering a fully functional game.

• 3D Game Engine (C++, OpenGL) [GitHub]

- Served as a Graphics Programmer and Producer in a 4-member team, designing and implementing core engine systems with a focus on rendering and graphics pipeline features.
- Built gameplay systems leveraging engine capabilities to ensure seamless engine-game integration, conducting playtesting to iterate on performance and user experience.

Course Projects – Machine Learning and Deep Learning

- Machine Learning 1 – Speech Emotion Recognition [GitHub]
  (Python, NumPy, Seaborn, Matplotlib)
  - Compared Random Forest and SVM classifiers for 6-class emotion recognition on CREMA-D (7,442 samples), achieving a macro-AUC of 0.82 (SVM) vs. 0.80 (RF) under class imbalance.

- Machine Learning 2 – Movie Recommender System [GitHub]
  (Python, NumPy, Seaborn, Matplotlib)
  - Co-developed a hybrid recommender on MovieLens 100K combining K-means++ clustered KNN, SVD matrix factorization, and knowledge-based filtering, reaching a best RMSE of 1.067.

- Deep Learning - CNN for Image Classification (TensorFlow/Keras, Google Colab) [GitHub]
  - Trained CNNs on CIFAR-10/100 and Animal-10 (~78% CIFAR-10 validation accuracy) and applied two-phase transfer learning with MobileNetV2.
  - Built a config-driven experiment pipeline with checkpointing, TensorBoard logging, early stopping, learning-rate scheduling, and class weighting to stabilize training under class imbalance.

- Deep Learning - Deep Reinforcement Learning – DQN Vacuum Agent (PyTorch) [GitHub]
  - Built a DQN from scratch with experience replay, target networks, and reward shaping, and scaled training across 5×5 to 15×15 grid environments.

PROFESSIONAL EXPERIENCE

Backstage Pass Institute of Gaming and Technology Jan 2019 – Aug 2024

Assistant Professor Hyderabad, India

- Mentored students in coursework and an incubator program, guiding projects from concept to product development.
- Taught game development using Unity3D and C#, enabling students to design and build interactive projects.
- Integrated technology into instruction and developed course materials to enhance engagement, critical thinking, and problem-solving.
- Provided individualized support, assessed performance, and delivered constructive feedback to improve student outcomes.

TapNext Studios Sep 2017 – Sep 2018

Game Developer Chennai, India

- Developed 5+ 2D and 3D mobile games for Android using Unity3D and C#, collaborating within a 5-member cross-functional team to ship production-ready titles.
- Designed and customized scalable game systems for 2+ sister companies, adapting to diverse product requirements and reducing integration time by 30% across projects.
- Built reusable gameplay frameworks and modular systems incorporating core gameplay mechanics and AI-driven behaviors, decreasing development time for new features by 30%.
- Managed the end-to-end game development pipeline from concept to production, delivering multiple titles on schedule and reducing production bottlenecks by 20%.

Nextwave Multimedia Aug 2016 – Sep 2017

Game Developer Hyderabad, India

Project: World Table Tennis Champs [Demo]

- Contributed to the development of a mobile 3D Table Tennis game from early concept through pre-launch using the Unity game engine, optimizing performance for resource-constrained devices.
- Implemented adaptive AI opponent behaviors with multiple difficulty levels, adjusting to player gameplay patterns to improve challenge and engagement.
- Built core gameplay modules, including tournament data generation and dynamic menu systems, applying Singleton and Object Pooling patterns to reduce memory overhead.
- Architected UI update systems using the MVC pattern to enforce separation of concerns and improve code maintainability, collaborating with design and art teams on a production-ready build.

SuperNova Games Oct 2015 – Sep 2016

Game Developer Hyderabad, India

Project: Cricket Dash (3D Mobile Infinite Runner) [Demo]

- Developed a 3D mobile infinite runner game using Unity3D and C#, serving as the sole programmer in a 2-member team alongside an artist and designer.
- Implemented core gameplay systems, including game logic, player mechanics, and progression systems.
- Leveraged object pooling to reduce memory overhead and maintain smooth real-time gameplay on mobile devices under increasing game pace.
- Architected a data-driven achievement system using JSON for configurable definitions and C# events/delegates to enable event-based tracking and real-time updates.

SKILLS

- Languages: C++, C#, Python, GLSL.
- Libraries: OpenGL, GLFW, GLAD, GLM, Assimp, FreeType.
- Game Engines: Unity (C#).
- Scripting & Tools: Lua, sol2, DOTween, Git, Visual Studio, CMake, RenderDoc.
- Machine Learning & Deep Learning: scikit-learn, TensorFlow/Keras, PyTorch, Pandas, NumPy, Matplotlib, Seaborn